DEEP KAYASTHA

AI / Machine Learning Developer

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ABOUT : Second-year AI/ML student at CHARUSAT with a track record of shipping production-grade, full-stack AI systems. Hands-on experience designing NLP pipelines, quantitative market intelligence platforms, and multilingual speech AI — from architecture through deployment. Strong team leader who coordinates cross-functional efforts under hackathon pressure.

PROJECTS

Armor.ai — Multilingual Financial Conversation Intelligence *2025*

- Built a 3-stage AI pipeline (STT → Translation → Insights) that processes audio in 14 Indian languages using Whisper Large V3 on GPU, with a tiered fallback to Faster-Whisper Medium/Tiny for reliability
- Engineered a 25-key JSON insights engine combining DistilBERT sentiment analysis, spaCy NER, a 50-keyword finance detector across 10 Indian scripts, and LLaMA 3.1 8B (NVIDIA NIM) for intent, risk, and summary extraction
- Designed full-stack architecture: React 19 frontend with Recharts analytics dashboard, Express + Supabase backend with JWT auth, and a Python AI module spawned as a subprocess — all connected via REST API
- Built for HACK2FUTURE 2.0 at IIIT Dharwad ; led AI pipeline design, NLP architecture, and backend integration for Team Lazy Legends

Tech Stack: *Python · PyTorch · Whisper Large V3 · DistilBERT · spaCy · LLaMA 3.1 8B (NVIDIA NIM) · React 19 · Node.js · Express · Supabase · Tailwind CSS*

TrendCaster — AI-Powered Indian Market Intelligence Platform *2025*

- Designed a 6-stage RAG pipeline: yfinance data fetch → proprietary signal engine → rolling PCA macro factor analysis → LLM input builder → ChromaDB/LlamaIndex vector store → Groq LLM for grounded answers
- Engineered 5 quantitative signals per instrument (trend strength, momentum acceleration, volatility regime, trend consistency, cycle position) across 14 tracked Indian market instruments including Nifty 50, Bank Nifty, and India VIX
- Built behavioural finance tools — Regret Simulator, Mistake Memory, Crowd vs Smart Money, Wait Power — giving retail investors data-backed decision support instead of noise
- Embedded 20 curated financial documents into the vector store and implemented intent-based query routing for contextually grounded answers

Tech Stack: *Python · Flask · React 19 · Vite · LangChain · ChromaDB · LlamaIndex · Groq LLM · MySQL · yfinance · scikit-learn · Tailwind CSS v4*

AEGIS-FIN — Real-Time Financial Risk Intelligence for SME Lending` *2025*

- Architected a multi-gateway risk intelligence platform integrating SME credit data, banking APIs, and live market feeds to serve India's underserved lending ecosystem
- Designed end-to-end data pipelines for financial risk scoring, applying ML models for default prediction and anomaly detection; built an interactive web dashboard for live visualization of risk signals
- Packaged the system into a full presentation deck and demo for hackathon evaluation; runner-up in the competition

Tech Stack: *Python · Pandas · Machine Learning · FastAPI · React · Tailwind CSS*

SKILLS

- **Languages:** Python, C++, C, JavaScript, Java, SQL
- **ML & Data:** Pandas, NumPy, Scikit-lear, TensorFlow, OpenCV, EDA, Feature Engineering, Model Evaluation
- **Web & Cloud:** React 19, Tailwind CSS, Flask, FastAPI, Express.js, Node.js, Supabase, MongoDB, Django, REST APIs
- **Tools:** Git, GitHub, VS Code, Streamlit, Jupyter, MySQL
- **Soft Skills:** Team leadership, workflow design, task distribution, cross-functional coordination under pressure

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning *2024 – 2028*

Charotar University of Science and Technology (CHARUSAT) · Student ID: 24AIML018 · 2nd Year

ACHIEVEMENTS & RECOGNITION

- Top 60 of 580+ teams across India — Hack2 Future Hackathon, IIIT Dharwad (national-level selection)
- 2nd Runner-Up — Markup Mania Hackathon
- 2nd Runner-Up — Debug Context Competition, AI/ML Department, CHARUSAT
- Kaggle: 9 upvotes on bank loan risk analysis notebook; 6 public notebooks spanning FinTech EDA, sector rotation, and traffic analysis